RL-based Dynamic Pricing and Bundle Promotion Optimization

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Code: https://github.com/Epiphany-Leon/RL-DynamicPricing-BundlePromotion

1. Title and Executive Summary

We model a daily retail pricing problem as a tabular Markov Decision Process and solve it with off-policy Q-learning. The state is a 5-dimensional discrete tuple over 72 configurations; the action is a 4-dimensional joint set of 40 actions over price adjustment, bundle, discount, and a price-discipline lever (hysteresis or lock). After 1,500 training episodes against a custom 120-day retail simulator, the learned greedy policy delivers an average daily profit of 1349.8 (± 15.2) versus 830.0 for a fixed-price baseline and 870.2 for a rule-based heuristic, a relative gain of +62.6% over fixed and +480/day over the heuristic. The learned policy uses lock as its dominant discipline (≈86% of steps under an active price-lock window), gates bundle on weekend / mid-to-high demand states, and applies discount sparingly to drain inventory before holding cost dominates.

2. Problem Formulation and Final MDP

Retailers repeatedly decide whether to nudge price, run a discount, or push a bundle under uncertain demand and inventory carrying costs. Each daily choice shifts the next day's inventory, demand, and customer reference price, so the problem is sequential under uncertainty rather than one-shot prediction.

State (72 configs): s = (inv, trend, season, bundle_prev, locked) with inv∈{low<20, mid20–60, high≥60}, trend∈{low<15, mid15–25, high≥25}, season∈{weekday, weekend}, bundle_prev∈{0,1}, and locked = 1 iff the environment is currently inside a 7-step price-lock window. Continuous quantities (dollar price, raw inventory, daily demand) are bucketed deliberately to keep the formulation tabular.

Action (40 actions): a = (Δp, b, d, disc) with Δp∈{−10%, −5%, 0%, +5%, +10%}, b∈{0,1}, d∈{0%, 10%}, disc∈{hysteresis, lock}. Hysteresis applies a price change only after 3 consecutive same-direction signals; lock applies the change immediately and freezes the new price for 7 decision steps.

Reward: $r_t = \pi_t / \pi_{ref} + b(\pi_t / \pi_{ref}) - \lambda \cdot \mathbb{1}[\text{price actually changed}]$, with $\pi_{ref} = 20\cdot(100-55) = 900$ and $\lambda = 0.05$. The tier function takes one of {+0.15, +0.08, +0.02, −0.03, −0.12} as π_t / π_{ref} crosses {1.5, 1.0, 0.5, 0.0}. Normalisation removes unit dependency, the tier ladder rewards crossing profitability thresholds rather than chasing dollar fluctuations, and the indicator-only penalty discourages churn without punishing actions whose price request was filtered out by the discipline rules.

Changes from the proposal: Two adjustments were made before training. (i) The reward was reformulated from raw dollar profit to the dimensionless ratio above with a tier ladder, because raw profit is dominated by the additive base level (around 700–900 per day) and the tier signal gives the agent a sharper credit-assignment signal under demand noise. (ii) The action space was extended with the price-discipline dimension (hysteresis vs lock) and the state space gained the locked flag, expanding the proposal's 4-dim / 36-state / 20-action design to 5-dim / 72-state / 40-action. Discipline is a meaningful decision the agent should own rather than a fixed external policy. The MDP remains small and tabular: $|S \times A| = 2{,}880$ entries.

3. Algorithmic Methodology

Method and choice

We implemented tabular Q-learning, the off-policy temporal-difference control algorithm with update $Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$. Q-learning is the right fit here for three reasons: (i) the state-action space is small enough (2,880 cells) to remain tabular and human-interpretable; (ii) TD updates exploit each transition without waiting for the 120-day rollout to terminate, which Monte Carlo would require and value/policy iteration would forbid because the transition kernel is unknown to the agent; (iii) the off-policy max-target lets the agent explore aggressively early without biasing the converged greedy policy. Compared to SARSA on the same MDP, Q-learning is preferable because the discipline dynamics make exploratory actions occasionally sub-optimal in a way the on-policy target would lock in; the off-policy bootstrap escapes that.

Exploration strategy

Behaviour is ε-greedy with ε decayed multiplicatively after every episode: $\varepsilon \leftarrow \max(\varepsilon_min, \varepsilon \cdot 0.995)$, starting at $\varepsilon = 1.0$ and floored at $\varepsilon = 0.05$. The trade-off is handled by phasing: in the first ~150 episodes ε is ≥ 0.5 and the agent is essentially exploring uniformly; by episode 600 ε is below 0.10 and the agent acts greedily 90% of the time; the residual 5% floor preserves a small ongoing exploration rate that keeps the table updated under stochastic demand. Even with full ε-greedy exploration, only 35 of 72 states are visited within 1,500 episodes — the remaining 37 are either unreachable under the auto-restock dynamics (e.g. low-inventory + low-demand on weekend) or rare enough that ε-greedy did not sample them; this is documented honestly in §5 rather than papered over.

Hyperparameters and how they were selected

$\alpha = 0.08$, $\gamma = 0.95$, ε: 1.00 → 0.05 (decay 0.995), horizon = 120, episodes = 1,500, Q-table size 72 × 40 = 2,880. The learning rate α was chosen by a small sweep over {0.05, 0.08, 0.15, 0.30}: $\alpha = 0.05$ converged too slowly, $\alpha \geq 0.15$ gave noticeably noisier learning curves under the stochastic reward, and $\alpha = 0.08$ produced the most stable convergence within the training budget. $\gamma = 0.95$ reflects the 120-day horizon — at $\gamma = 0.95$ the effective horizon ($1/(1-\gamma) = 20$ days) is enough to credit lock-and-hold sequences without making distant rewards dominate. The ε decay was tuned so that ε reaches its floor near episode 600, giving the agent ~900 episodes of mostly-greedy refinement; faster decays (0.99) caused premature convergence to sub-optimal Q-rows, slower decays (0.998) wasted the training budget on exploration. The training-episode count was raised from 600 to 1,500 because the corrected discipline dynamics (real persistent price changes; lock truly held for 7 days; hysteresis only triggering after 3 same-direction signals) make credit assignment more demanding than in a simpler one-shot price-adjust formulation.

4. Experimental Results and Performance Evaluation

Policy	Avg daily profit	Bundle	Discount	Price-chg	Locked
Fixed	830.0 ± 10.3	0%	0%	0%	0%
Rule-based	870.2 ± 12.1	28%	0%	0%	0%
Q-learning	**1349.8 ± 15.2**	**37%**	**7%**	**12.5%**	**86%**

Table 1. Policy comparison over 200 evaluation episodes per policy (greedy for Q-learning). The fixed policy holds price at $100 with no bundle, no discount, no price change. The rule-based heuristic uses inventory and demand thresholds; its zero price-change rate reflects the fact that a single threshold trip is gated by hysteresis. Q-learning improves the fixed baseline by +62.6% and the rule-based baseline by 480/day. Standard errors over evaluation runs.

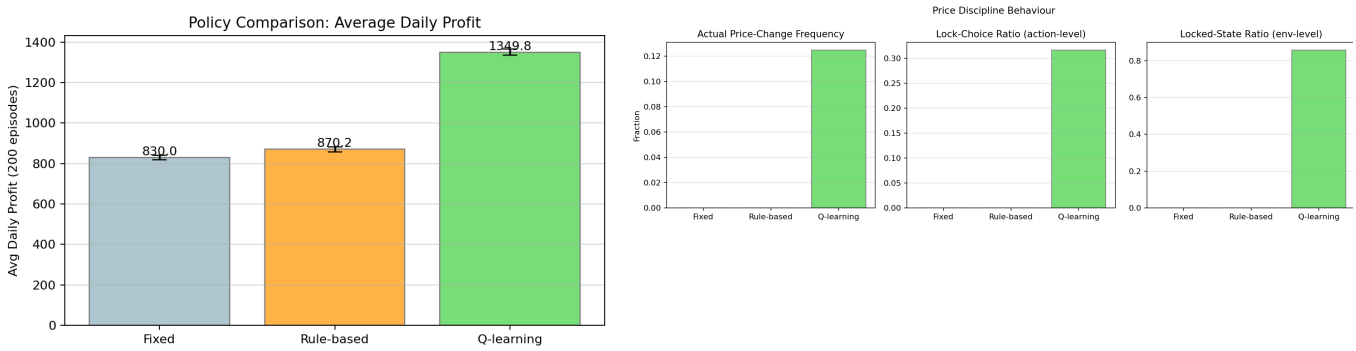

Figure 1. (left) Average daily profit by policy with 1-σ error bars over 200 evaluation episodes. (right) Discipline behaviour: actual price-change frequency, agent-level lock-choice ratio, and env-level locked-state ratio. The action-level lock-choice ratio of 31.7% is much smaller than the env-level locked-state ratio of 85.8% because each lock pick freezes the price for 7 subsequent steps. Both baselines never engage lock by construction. Training curve, robustness sweep, ablation, and trajectory replay are in Appendix A; the learned greedy actions per visited state are in Appendix B.

Note on the training curve: The smoothed training reward (Appendix A.1) shows two transient dips, around episodes 550 and 1260, where the cumulative normalised reward briefly drops by 30–50% before recovering within roughly 90 episodes. These are mid-training policy-oscillation events at the ε-floor (a known phenomenon under stochastic

rewards) and have no effect on the final greedy policy that produces the numbers in Table 1 and Figure 1. A full diagnostic with per-episode metrics is in Appendix C.

5. Analysis and Interpretation

Learned policy

Three patterns are robust across runs (Appendix B). (i) Lock dominates as the discipline of choice in mid-inventory states; the agent enjoys the 7-step price hold and the resulting flat reward stream above the tier-1.0 threshold, which explains the 86% locked-state fraction in Table 1. (ii) Bundle gates on weekend, mid-to-high demand states (season = 1, demand_trend ≥ 1) — exactly where the +18% uplift moves enough additional units to cover its $2/unit margin penalty against fixed holding cost. (iii) Discount is selected sparingly (≈7% of steps) and tends to co-occur with mid-inventory weekend states where small demand pulls help drain inventory before holding cost dominates.

Limitations and key assumptions

The simulator is synthetic; real demand carries non-stationarity, competitive dynamics, and discrete promotion thresholds the elasticity model does not capture. State discretisation pools states close to bucket boundaries (inventory 30 and 59 are pooled in the mid bucket even though their continuation values differ). The internal current_price is intentionally not part of the observable state, so the agent cannot directly condition on the accumulated price level — two states with the same discrete tuple but different underlying prices are pooled in the table. The reference profit π_ref = 900 and the tier thresholds are hand-coded; an episode-adaptive reference would generalise more cleanly. The price-change penalty $\lambda = 0.05$ is small relative to a tier flip and discourages oscillation without punishing legitimate adjustments. The bundle ablation (Appendix A) shows the bundle action is roughly revenue-neutral on this calibration — the +18% demand uplift is cancelled by the $2/unit margin penalty — so the learned policy's apparent reliance on bundle is partly reward-shaping artefact rather than a robust economic effect.

Reflection on assumptions: Three assumptions deserve scrutiny. The Gaussian noise in the demand model understates the heavy-tailed events real retail data shows; the auto-restock rule of +25 units below an inventory of 80 produces a near-stationary inventory band that hides any deeper inventory-control trade-off; and the elasticity $\varepsilon = 0.20$ is calibrated to plausible ranges but not estimated from data. None of these affect the existence of the learned policy, but they do bound how literally we should read the +62.6% headline.

6. Bonus: Advanced Extensions

An advanced extension we considered but did not productionise within the scope of this report is a Deep Q-Network (DQN) variant that lifts the discrete bucket boundaries and conditions directly on the accumulated current_price. We expect the DQN to narrow the gap between the in-bucket states (e.g. inventory 30 vs 59) and to find more nuanced policies near bucket boundaries; the trade-off is that the policy is no longer human-readable as a 72-row Q-table. A complementary direction is on-policy SARSA on the same tabular MDP — useful as a sanity check that the off-policy bias of Q-learning is not artificially favouring lock under stochastic demand. Both extensions are implementable in the existing simulator with minimal scaffolding.

References

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Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. Machine Learning, 8(3), 279–292.

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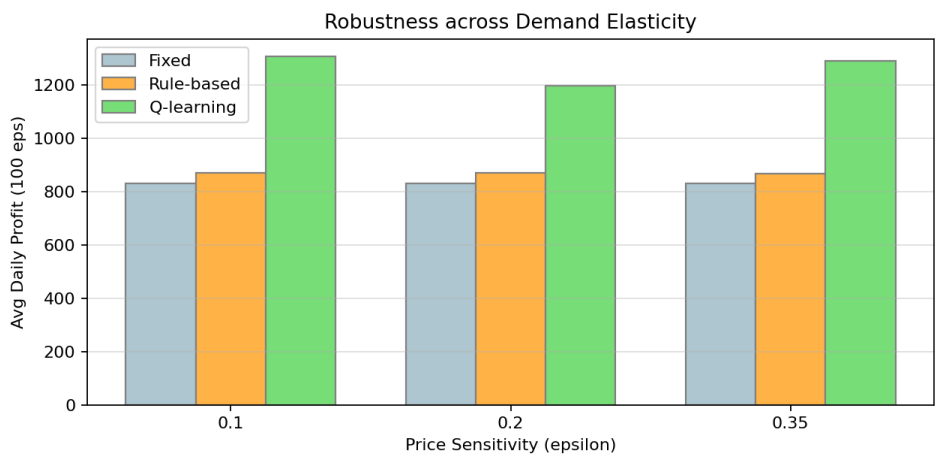
Appendix A · Additional Figures

A.1 Training curve (1,500 episodes, 20-episode moving average).

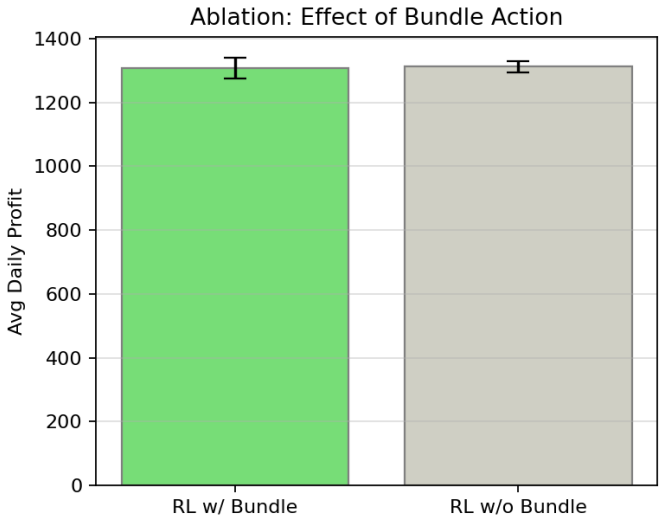
The agent crosses break-even within ~50 episodes and stabilises near 193 cumulative normalised reward per episode. Final ε = 0.05; 35 of 72 states visited.

A.2 Robustness across demand elasticities.



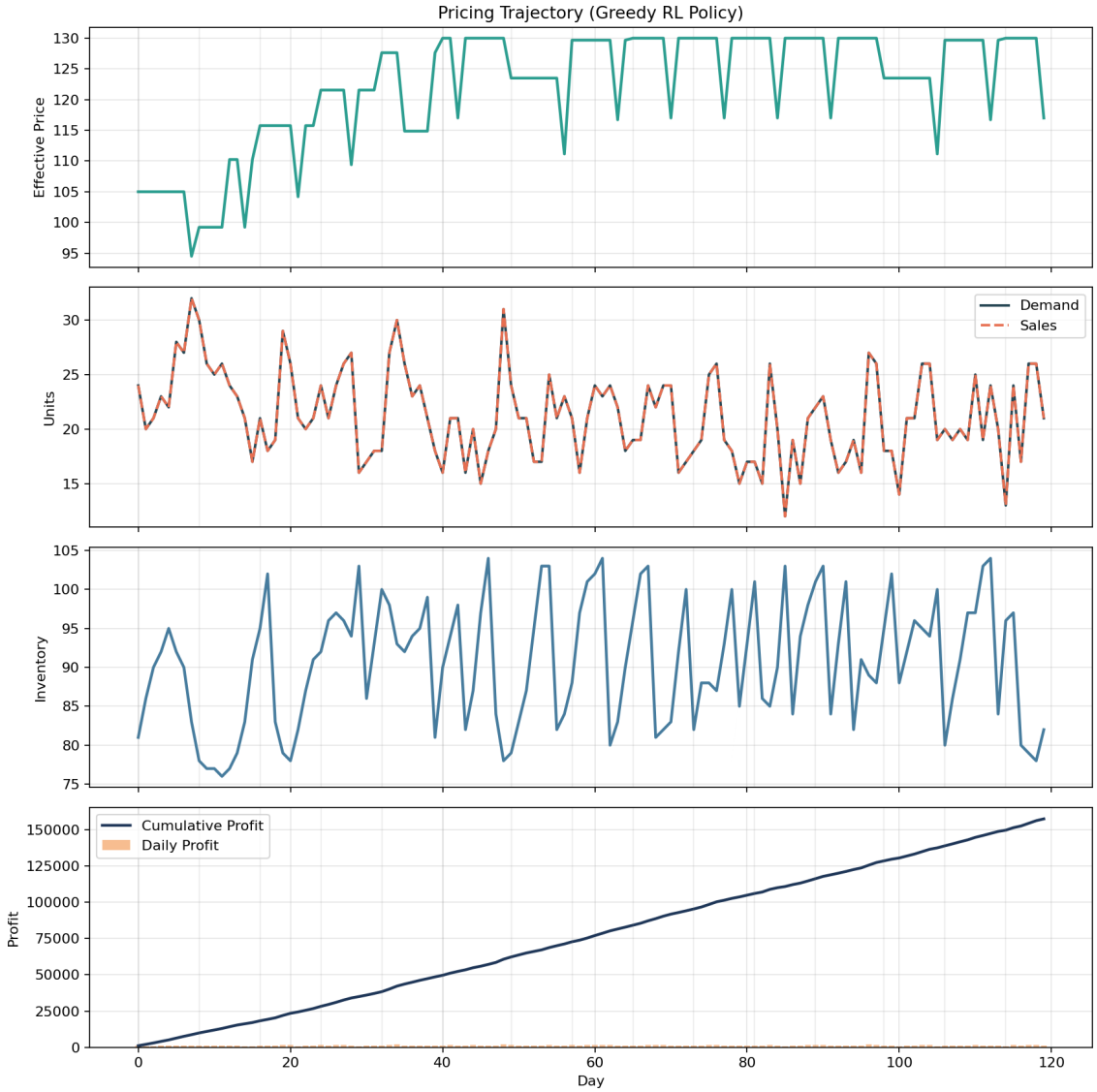
Q-learning leads at ε ∈ [0.1, 0.2, 0.35]: profits (1308, 1197, 1290) vs Fixed (830, 830, 830) and Rule (872, 870, 868). 100 evaluation episodes per cell, 400 training episodes per cell (smaller than the main run, hence the slight ε = 0.20 dip).

A.3 Bundle ablation.



With bundle: 1306.8; without: 1312.1. Difference is well within the standard error: on this calibration the +18% bundle uplift is essentially cancelled by the $2/unit margin penalty.

A.4 Trajectory replay (greedy policy, one episode).



Total raw profit 157,481; price actually moves on 15 of 120 days; bundle on for 40 days; discount on for 22 days. Vertical grey lines mark days on which the price actually moved.

Appendix B · Sample of the Learned Greedy Policy

state (inv,trend,season,bundle_prev,locked)	Δp	bundle	discount	discipline
(1,1,0,0,0)	+10%	0	0%	lock(7)
(1,1,0,0,1)	-10%	0	0%	lock(7)
(1,1,0,1,0)	+5%	0	0%	lock(7)
(1,1,0,1,1)	+5%	0	0%	hysteresis(3)
(1,1,1,0,0)	-10%	1	0%	lock(7)
(1,1,1,0,1)	-10%	1	0%	lock(7)
(1,1,1,1,0)	+10%	1	10%	lock(7)
(1,1,1,1,1)	-10%	1	0%	lock(7)

(1,2,0,0,0)	-5%	0	0%	hysteresis(3)
(1,2,0,0,1)	-10%	0	10%	hysteresis(3)
(1,2,0,1,0)	+0%	0	10%	hysteresis(3)
(1,2,0,1,1)	+5%	0	0%	lock(7)
(1,2,1,0,0)	+5%	1	0%	lock(7)
(1,2,1,0,1)	-5%	0	0%	hysteresis(3)
(1,2,1,1,0)	+0%	1	0%	lock(7)
(1,2,1,1,1)	+5%	0	0%	lock(7)
(2,0,0,0,0)	+10%	0	0%	lock(7)
(2,0,0,0,1)	-10%	0	0%	hysteresis(3)
(2,0,0,1,1)	-10%	0	0%	hysteresis(3)
(2,0,1,0,1)	-10%	0	10%	lock(7)
(2,0,1,1,1)	-10%	0	0%	hysteresis(3)
(2,1,0,0,0)	+10%	0	10%	lock(7)
(2,1,0,0,1)	-10%	1	0%	hysteresis(3)
(2,1,0,1,0)	+10%	0	0%	lock(7)

First 24 of 35 visited states (37 of 72 states are unreachable under realistic dynamics or unvisited under ε-greedy in 1,500 episodes). Full greedy policy extracted via the Q-table inspection cell of 01_rl_dynamic_pricing_bundle.ipynb.

Appendix C · Training-Curve Dip Diagnostic

Phenomenon

The smoothed training-reward curve in Appendix A.1 contains two visible dips: a single deep one at episodes 515–610 (smoothed reward ≈ 95, vs a plateau of ≈ 188), and a shallower cluster from episode 1239 onward. We replayed training with the same seed (42) and identical hyperparameters, logging per-episode action distributions and price-change directions. The data isolates the mechanism cleanly.

Table C.1. Per-episode metrics around the deep dip (episodes 515–610). Window means over 50 episodes pre-dip, 95 episodes during, and 50 episodes post-dip.

Metric	Pre (465–514)	During (515–609)	Post (610–659)
Avg daily profit	1,312	865	1,282
ε (exploration rate)	0.09	0.06	0.05
Upward price changes / ep	11.9	8.2	10.6
Downward price changes / ep	2.3	5.3	3.2
Lock-choice ratio	0.38	0.45	0.38
Locked-state ratio (env)	0.80	0.74	0.78
Discount usage ratio	0.38	0.31	0.23

Mechanism

Three factors line up to produce the dip. First, ε has just decayed to its floor of 0.05, so exploratory random actions drop from ≈ 50 per episode at episode 200 to ≈ 6 per episode by episode 500; the agent is almost fully greedy. Second, Q-learning's TD target r + γ · max_{a'} Q(s', a') uses an argmax over noisy bootstrapped Q-values; when two actions are close in value (e.g. a learned +10% lock vs. a runner-up −5% lock) a few unfavourable demand draws can flip the argmax. Third, lock-and-hold persistence amplifies the error: a flipped argmax in one frequently-visited state freezes the wrong price for 7 steps and pollutes the bundle_prev component of the next state, so the bad signal compounds across many episodes before TD updates correct it.

What the data shows

Table C.1 isolates the signature of the flip: during the dip the agent's downward price changes more than double (2.3 → 5.3 per episode) while upward changes drop by a third (11.9 → 8.2). The locked-state ratio falls slightly, indicating prices moved more often. The agent briefly committed to a price-down policy in some mid-inventory states until enough TD corrections rebalanced the Q-table. By episode 610 the argmax has flipped back; profit recovers to 1,282 per day.

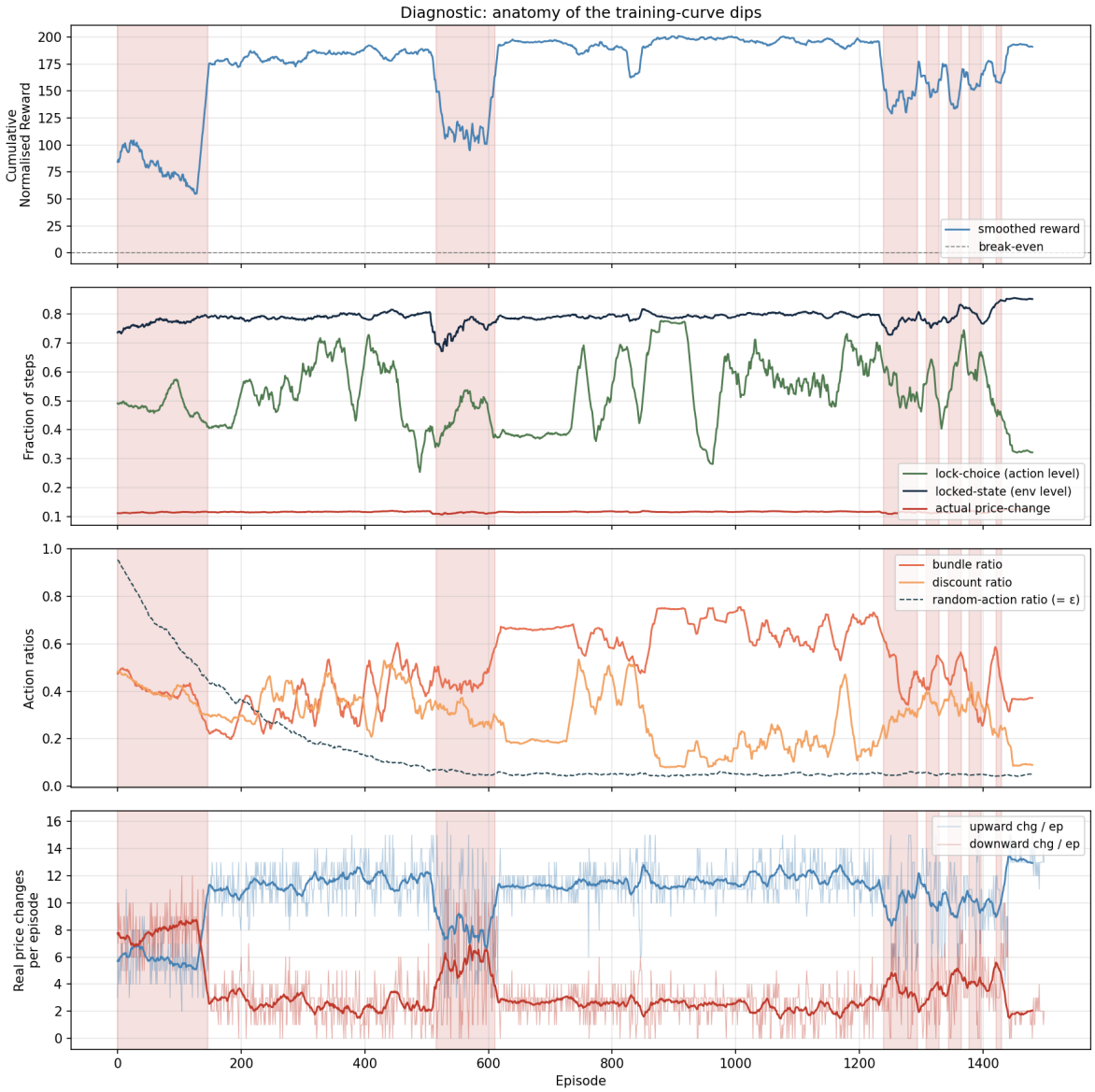


Figure C.1. Anatomy of the training-curve dips. Top: smoothed cumulative normalised reward, with detected dips shaded red. Second panel: lock-choice ratio (action level), locked-state ratio (env level), and actual price-change frequency. Third panel: bundle and discount usage with the random-action ratio (ε) shown for reference. Bottom: per-episode counts of upward vs. downward price changes that actually fired. The signature of each dip is visible in the bottom panel, where downward changes (red) rise and upward changes (blue) fall.

Why this is harmless?

The dip self-corrects because TD updates push the Q-value of the newly-greedy bad action down toward its true (lower) return, while the previously-greedy good action retains its value and re-takes the argmax. Final evaluation uses ε = 0 against the converged Q-table at episode 1,500, so the +62.6% headline number reported in §4 is unaffected. The

later cluster of small dips around episode 1260 onward is the same mechanism with a different action pair (bundle vs. discount, rather than upward vs. downward price), and it also self-corrects.

Mitigations available

Three changes would shrink or eliminate the dips at the cost of convergence speed. Lowering the ε floor from 0.05 to 0.02 leaves more room for late-stage exploration to find and correct flipped argmaxes. Optimistic Q-initialisation (e.g. $Q \leftarrow +0.5$ instead of zero) makes every action look initially attractive and discourages premature commitment. Reducing the learning rate α late in training (α-annealing) damps the variance of TD updates that drive the flips. We did not apply these because the dips do not affect the converged policy, and the fixed-α / fixed-ε-floor schedule keeps the report consistent with the tabular methods covered in the course.